

EASA COLLEGE
OF ENGINEERING & TECHNOLOGY (ECET)
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# DEVELOP A HYPERPARAMETER-TUNING SYSTEM (GRID SEARCH OR RANDOM SEARCH) FOR A TEXT CLASSIFIER AND REPORT THE OPTIMAL PARAMETERS FOUND

A MINI PROJECT REPORT

Submitted by

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Register No: E720524AM014

*in partial fulfillment for the award of the degree of*

**BACHELOR OF ENGINEERING**

IN

**COMPUTER SCIENCE AND ENGINEERING**

**(ARTIFICIAL INTELLIGENCE & MACHINE LEARNING)**

EASA COLLEGE OF ENGINEERING & TECHNOLOGY

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OCTOBER 2026

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## BONAFIDE CERTIFICATE

Certified that this mini project report "DEVELOP A HYPERPARAMETER-TUNING SYSTEM (GRID SEARCH OR RANDOM SEARCH) FOR A TEXT CLASSIFIER AND REPORT THE OPTIMAL PARAMETERS FOUND" is the bonafide work of "NAVANEETHA.T" who carried out the mini project work under my supervision.


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# Abstract

This project develops a hyperparameter-tuning system for a text classifier and reports the optimal parameters found. A scikit-learn pipeline combining a TF-IDF vectorizer and a LinearSVC classifier was trained on four categories of the 20 Newsgroups dataset (3,899 documents). Both Grid Search (36 candidate settings) and Random Search (20 sampled settings) were run with 3-fold cross-validation and macro F1-score as the objective. Grid Search selected C = 0.1, max_df = 0.75, min_df = 1 and ngram_range = (1, 2) with a cross-validation F1 of 0.9118. Random Search reached a similar cross-validation F1 of 0.9099 in 83.6 seconds compared with 133.8 seconds for Grid Search. On the held-out test set, all models scored about 92 to 93 percent accuracy, and the untuned baseline performed slightly better than the tuned models, showing that the default configuration was already close to optimal for this task.

# 1. Introduction

Text classification assigns documents to predefined categories. The performance of a classifier depends strongly on hyperparameters such as the regularization strength and the feature-extraction settings, which are not learned from data and must be chosen by the practitioner. Manual selection is slow and unreliable, so automated search methods are used. This project implements and compares two common strategies, Grid Search and Random Search, and reports the best configuration found.

**Objectives:** (1) build a leakage-free text classification pipeline; (2) implement Grid Search and Random Search; (3) report the optimal parameters; (4) compare the methods against each other and against an untuned baseline.

# 2. Methodology

## 2.1 Dataset

The 20 Newsgroups dataset was loaded with scikit-learn using four categories: comp.graphics, rec.sport.hockey, sci.space and talk.politics.mideast. Headers, footers and quoted replies were removed to avoid metadata leakage. The 3,899 documents were split 80/20 into 3,119 training and 780 test documents with stratification. The test set was used only once, for final evaluation.

## 2.2 Pipeline

A scikit-learn Pipeline joins a TfidfVectorizer (lowercasing and English stop-word removal) and a LinearSVC classifier. Because vectorization is inside the pipeline, it is re-fitted within each cross-validation fold, which prevents data leakage.

## 2.3 Search Spaces

| Parameter | Grid Search | Random Search |
|---|---|---|
| tfidf__max_df | 0.75, 0.9, 1.0 | 0.6, 0.75, 0.9, 1.0 |
| tfidf__min_df | 1, 2 | 1, 2, 3, 5 |

| | | |
|---|---|---|
| tfidf__ngram_range | (1,1), (1,2) | (1,1), (1,2), (1,3) |
| clf__C | 0.1, 1, 10 | log-uniform(0.01, 100) |
| Candidates | 36 | 20 (sampled) |

## 2.4 Evaluation Setup

Both searches used 3-fold cross-validation with macro F1-score as the scoring metric (108 fits for Grid Search, 60 fits for Random Search). The best estimator from each search was then evaluated on the test set using accuracy, macro F1, a classification report and a confusion matrix. An untuned pipeline with default parameters served as the baseline.

# 3. Experimental Results

## 3.1 Optimal Parameters

| Parameter | Grid Search | Random Search |
|---|---|---|
| clf__C | 0.1 | 5.408 |
| tfidf__max_df | 0.75 | 1.0 |
| tfidf__min_df | 1 | 1 |
| tfidf__ngram_range | (1, 2) | (1, 3) |
| Best CV macro F1 | 0.9118 | 0.9099 |

## 3.2 Comparison of Methods

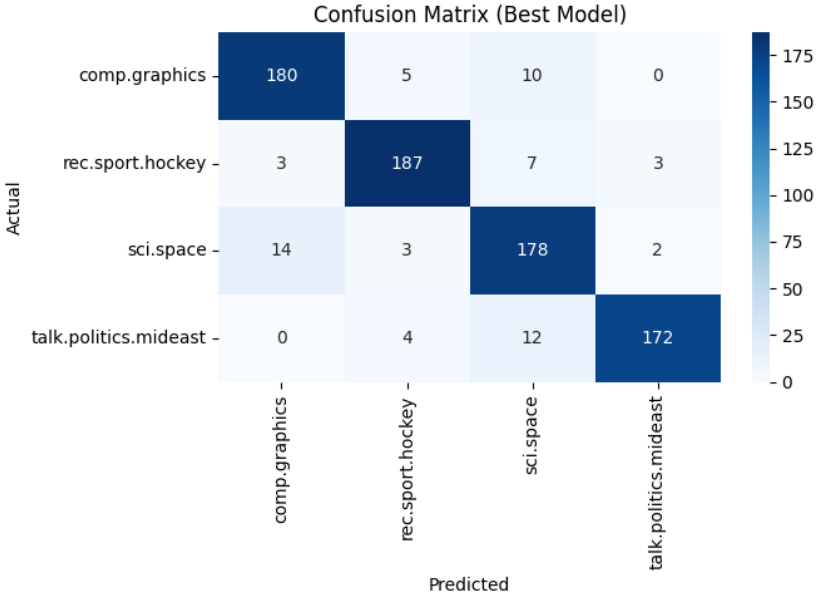
| Method | Candidates | Best CV F1 | Test Acc. | Test Macro-F1 | Time (s) |
|---|---|---|---|---|---|
| Grid Search | 36 | 0.9118 | 0.9192 | 0.9198 | 133.8 |
| Random Search | 20 | 0.9099 | 0.9256 | 0.9260 | 83.6 |
| Baseline (no tuning) | 1 | - | 0.9269 | 0.9278 | - |

Grid Search obtained the highest cross-validation score, so its configuration was selected as the final model according to the cross-validation criterion. Random Search came within 0.002 of this score while using about 38 percent less time.

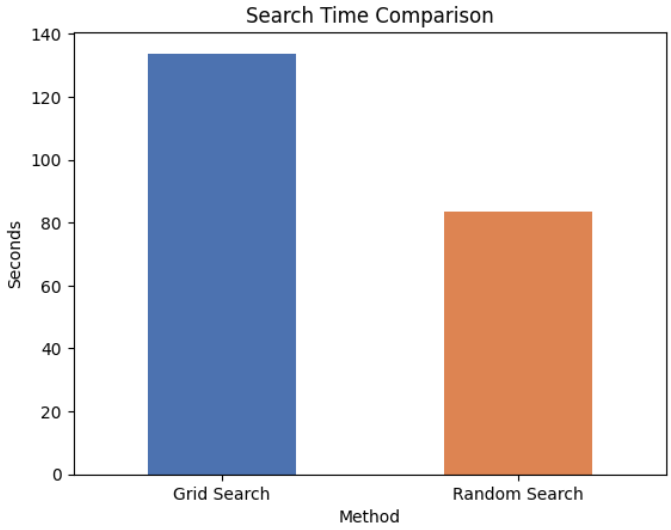
## 3.3 Final Model Performance (Grid Search best model)

| Class | Precision | Recall | F1 | Support |
|---|---|---|---|---|
| comp.graphics | 0.91 | 0.92 | 0.92 | 195 |
| rec.sport.hockey | 0.94 | 0.94 | 0.94 | 200 |
| sci.space | 0.86 | 0.90 | 0.88 | 197 |
| talk.politics.mideast | 0.97 | 0.91 | 0.94 | 188 |
| Accuracy | | | 0.92 | 780 |

## 3.4 Confusion Matrix




## 3.5 Search Time Comparison



# 4. Discussion

Cross-validation scores for the two searches were nearly identical (0.9118 and 0.9099), showing that Random Search can find competitive settings with fewer evaluations. Both methods favoured n-gram features beyond unigrams, and Grid Search preferred strong regularization (C = 0.1).

On the test set, the untuned baseline (accuracy 0.9269) matched or slightly exceeded the tuned models (0.9192 and 0.9256). The differences are under one percentage point and are within normal sampling variation for a test set of 780 documents. This indicates that LinearSVC with default settings is already a strong choice for this dataset, and that tuning gave limited additional benefit. The weakest class was sci.space (F1 = 0.88), which is likely confused with comp.graphics because of overlapping technical vocabulary.

# 5. Conclusion

A complete hyperparameter-tuning system was built using Grid Search and Random Search on a TF-IDF and LinearSVC text classification pipeline. The optimal parameters found by Grid Search were C = 0.1, max_df = 0.75, min_df = 1 and ngram_range = (1, 2), with a cross-validation macro F1 of 0.9118 and a test accuracy of 0.919. Random Search was faster and nearly as accurate. The tuned models did not clearly outperform the default baseline on the test set, which is a useful finding in itself: tuning should always be compared with a baseline. Future work could explore wider search spaces, other classifiers such as Logistic Regression or Random Forest, and Bayesian optimization.

# 6. References

1. Pedregosa et al., Scikit-learn: Machine Learning in Python, JMLR 12, 2011.
2. Bergstra and Bengio, Random Search for Hyper-Parameter Optimization, JMLR 13, 2012.
3. Scikit-learn documentation: GridSearchCV, RandomizedSearchCV, 20 Newsgroups dataset.